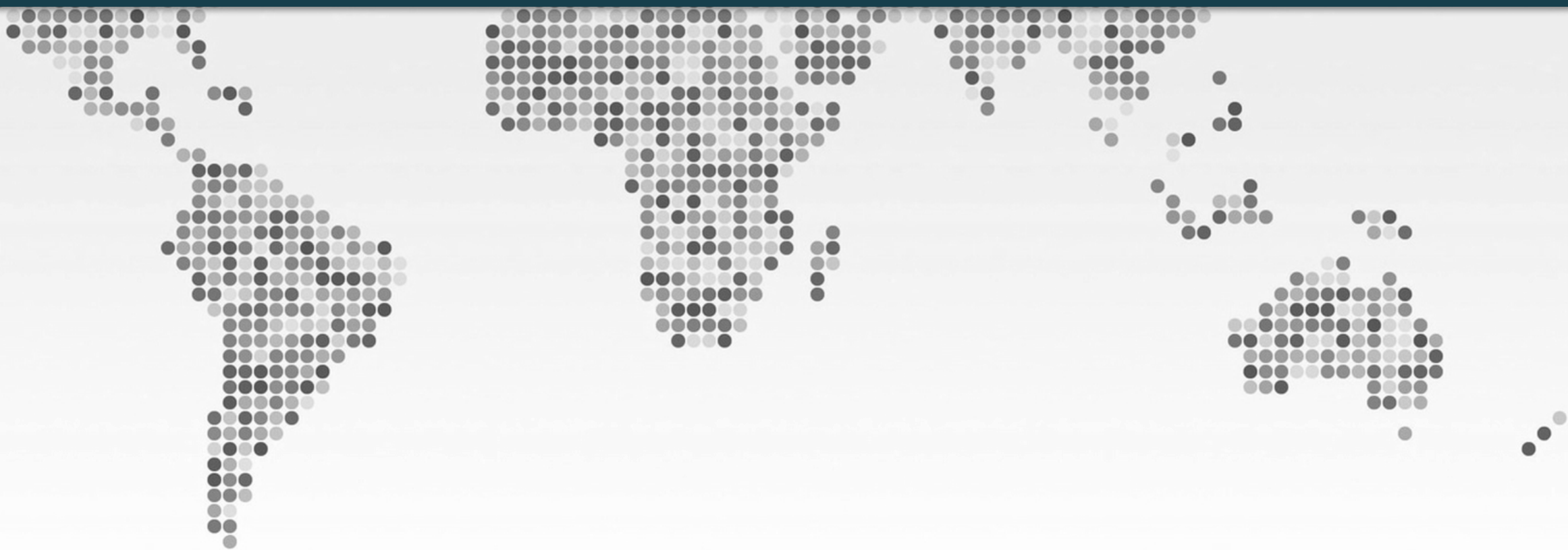


Software Engineering for Artificial Intelligence

Sustainability of AI-Enabled Systems



Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: What It Is About

Sustainability. Meeting the needs of the present without compromising the ability of future generations to meet their own needs.

In other words, sustainability is about maintaining balance and continuity, using resources wisely and responsibly, considering both present-day and future impacts.

Social sustainability

Focus on societal communities and the factors that erode trust in society. This includes social equity, justice, etc.

Economic sustainability

Focus on assets, capital, and added value. This includes prosperity, investments, etc.

Environmental sustainability

Focus on the long-term effects of human activities on natural systems. This includes climate change, food production, etc.

Individual sustainability

Focus on the well-being of humans as individuals. This includes mental and physical well-being, education, self-respect, etc.

Technical sustainability

Focus on the longevity and the evolution of information, systems, and infrastructure. It includes maintenance, innovation, etc.

Importantly, these are *interdependent*, not independent. As such, each dimension impacts on the other. This makes sustainability an optimization problem that must be solved by considering the different trade-offs the each design choice may imply.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: What It Is About

Sustainability. Meeting the needs of the present without compromising the ability of future generations to meet their own needs.

For instance, let's consider a company that generates novels using a large language model and charges readers a flat monthly fee to read as much as they want.

Social sustainability

Does it replace human authors? Does it change cultural production? Does it promote inclusive access to content?

Economic sustainability

Can the company sustain infrastructure costs? Are subscriptions affordable for users?

Environmental sustainability

What about the energy cost of training (periodic) and inference (continuous)?

Individual sustainability

What is the impact on job displacement/augmentation? Would generated novels engage readers and promote education?

Technical sustainability

What about model maintenance? What about the scalability of serving infrastructure?

If we lower the subscription price, we will likely have more users; but more users would increase inference cost, impact scalability, and increase environmental costs. Similarly, switching to a larger model would improve quality, but increase costs and energy impact.

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To a large extent, non-functional requirements optimization is, at its core, a form of sustainability engineering.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

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Sustainability. Meeting the needs of the present without compromising the ability of future generations to meet their own needs.

In addition to the concept of stratification, sustainability in software engineering is defined by other paradigms that help us determine its nature.

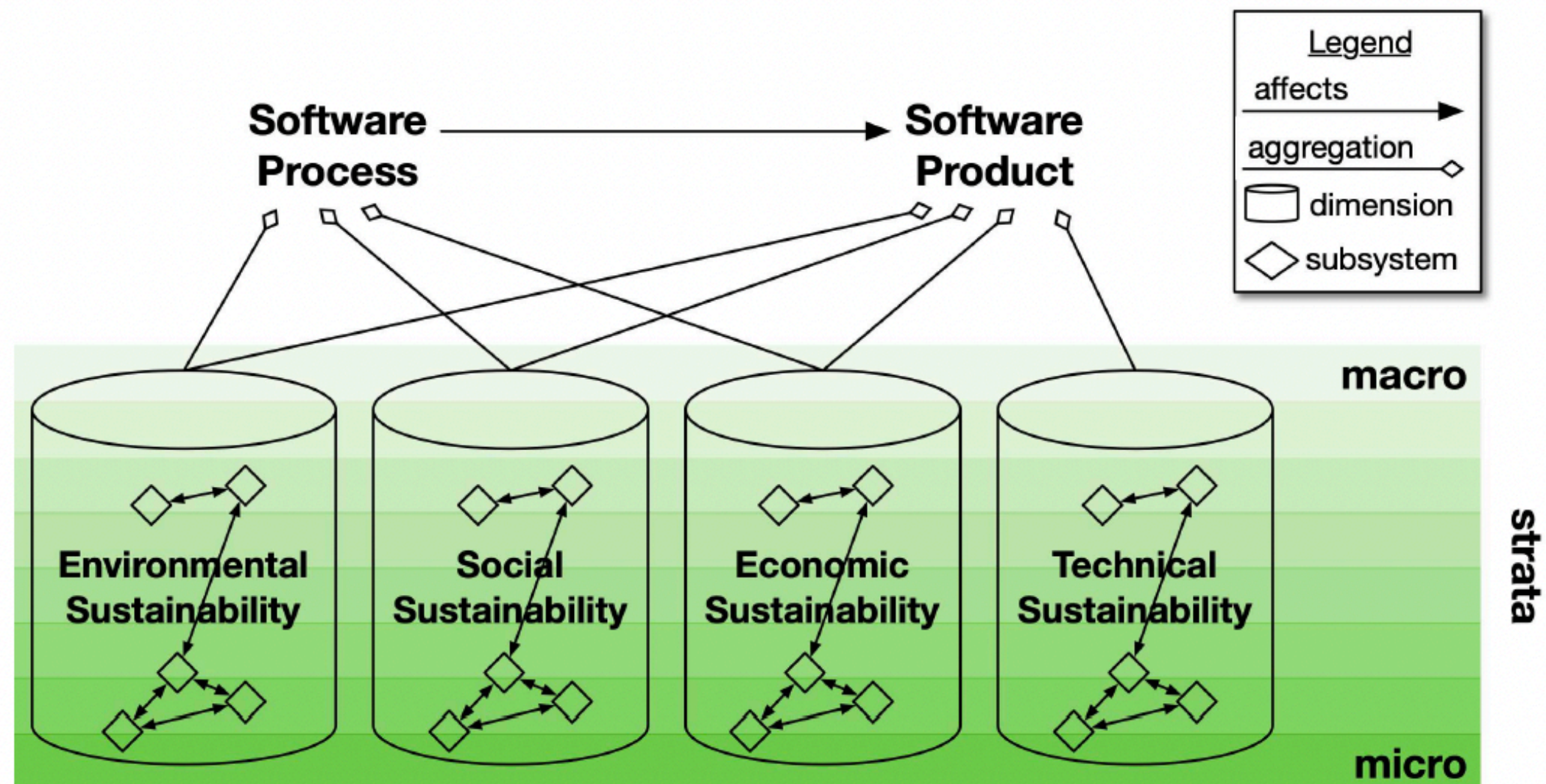
- (1) *Sustainability is multisystemic*: It does not come from one system but emerges from interactions among multiple systems (technical, social, etc.). Thinking in terms of dimensions hides how these systems interact, while we need to understand sustainability as something that arises from system interdependencies.
- (2) *Process Sustainability vs. Product Sustainability*: Software engineering affects sustainability both through the development process (e.g., how teams work) and the product (e.g., long-term usability). The practices are not always aligned: we can have a sustainable process but an unsustainable product and vice versa.
- (3) *Sustainability is Participatory*: Sustainability must involve multiple stakeholders because it is not a task only for developers. Sustainability may involve people from diverse user groups, including those with less technological experience and differing support needs.

Based on these paradigms, we can again rethink sustainability as an integrated concept throughout the software development lifecycle.

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Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: Challenges of Classical AI through a Sustainability Lens

Social sustainability

Focus on societal communities and the factors that erode trust in society. This includes social equity, justice, etc.

Ethics and fairness in AI are core components of social sustainability, as they directly affect trust, equity, and societal well-being.

Yet, equating social sustainability to fairness would be erroneous. Fairness is about whether a system treats people correctly. Social sustainability is about whether the system contributes to a **healthy society over time**.

Trust. The degree to which individuals and society have confidence that an AI system behaves reliably, transparently, and in alignment with expected norms and values.

Impact on social structures. The extent to which an AI system reshapes economic, professional, and institutional arrangements within society.

Behavioral and societal effects. The influence of AI systems on individual behaviors and collective societal dynamics over time.

Inclusion and participation. The degree to which diverse stakeholders are represented in data, involved in system design, and able to benefit from the system.

These dimensions determine whether an AI system contributes to a society that is fair, inclusive, and resilient over time.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: Challenges of Classical AI through a Sustainability Lens

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From an organizational perspective, this involves defining **accountability structures** (who is responsible for AI decisions), establishing **AI governance policies** (e.g., risk classification, audits), and promoting a **culture of transparency and documentation**.

From an engineering perspective, this involves **logging and traceability** of decisions, **explainability** techniques, **testing** for robustness and failure cases, and **monitoring** in production.

In other terms, trust is not really an abstract concept, but a socio-technical decision that involves governance and engineering - and that basically involves the instantiation of the methods we discussed in this course.

Trust. The degree to which individuals and organizations behave reliably, transparently, and responsibly.

Impact on social structures. The effect of AI on professional, and institutional arrangements.

Behavioral and societal effects. The effect of AI on collective societal dynamics over time.

Inclusion and participation. The degree to which data, involved in system design, is inclusive and participatory.

These dimensions determine whether a system is sustainable, inclusive, and resilient over time.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

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From an organizational perspective, this involves **defining impact assessments** before deployment and **planning for reskilling/upskilling** of affected roles.

From an engineering perspective, this involves designing systems to **augment, not replace** humans when possible, e.g., human-in-the-loop and interactive design instruments.

Again, this is not abstract, but a socio-technical decision.

Trust. The degree to which individuals and organizations behave reliably, transparently, and honestly.

Impact on social structures. The effect of AI on professional, and institutional arrangements.

Behavioral and societal effects. The effect of AI on collective societal dynamics over time.

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From an organizational perspective, this involves defining **acceptable use policies**, conducting **longitudinal studies** on user behavior, and including **ethics reviews** in the lifecycle.

From an engineering perspective, this involves planning **A/B testing** with behavioral metrics, designing for **human-in-the-loop** interaction, or **rate-limiting** or **safeguards** for misuse.

Again, this is not abstract, but a socio-technical decision.

Trust. The degree to which individuals and organizations behave reliably, transparently, and honestly.

Impact on social structures. The degree to which AI systems affect professional, and institutional and social structures.

Behavioral and societal effects. The degree to which AI systems affect collective societal dynamics over time.

Inclusion and participation. The degree to which individuals and organizations are included, involved in system design, and participate in the system.

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From an organizational perspective, this involves **diverse stakeholders** in requirements engineering, and ensuring **accessibility policies**.

From an engineering perspective, this involves planning **dataset auditing** for representation gaps, **fairness-aware** ML techniques, **accessibility-by-design**, and **participatory design approaches**.

Again, this is not abstract, but a socio-technical decision.

Trust. The degree to which individuals and organizations behave reliably, transparently, and honestly.

Impact on social structures. The degree to which AI systems affect professional, and institutional arrangements.

Behavioral and societal effects. The degree to which AI systems affect collective societal dynamics over time.

Inclusion and participation. The degree to which data, involved in system design, reflects the diversity of the population.

These dimensions determine whether a system is inclusive, and resilient over time.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: Challenges of Classical AI through a Sustainability Lens

Economic sustainability

Focus on assets, capital, and added value. This includes prosperity, investments, etc.

Economic sustainability may depend high cost of data collection/labeling, expensive hardware (GPUs, TPUs), and iterative experimentation cycles.

These issues relate to economic sustainability, as they can create significant barriers to market entry, effectively limiting competition to large organizations.

In addition, it is worth remarking that AI systems often incur **continuous costs**, not one-time investments, e.g., ongoing inference costs, maintenance, or scaling with user base. As such, **revenue must scale with operational cost**.

From an organizational perspective, companies should evaluate **build vs buy vs reuse choices** to ensure long-term economic viability and avoid unnecessary costs or dependencies.

Investing in internal skills, data management, and tooling helps **reduce reliance on external providers** and improves long-term competitiveness.

As AI systems incur ongoing costs, organizations need to actively **monitor usage and enforce cost controls**, like budgets, usage limits, and efficiency metrics, to prevent unsustainable expenses.

In other terms, keeping economic sustainability under control requires organizations to have **management and governance structures** through which costs are monitored and controlled, and to invest in skill development to reduce dependency on external solutions.

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In addition, it is worth remarking that AI systems often incur **continuous costs**, not one-time investments, e.g., ongoing inference costs, maintenance, or scaling with user base. As such, **revenue must scale with operational cost**.

From an engineering perspective, **improving data quality and relevance can often yield better results than increasing model complexity**, reducing both development and operational costs.

Engineers should treat **cost as a first-class requirement**, balancing performance with resource consumption when selecting models and architectures.

Automating and reusing pipelines minimizes redundant work, reduces experimentation costs, and improves the efficiency of the AI lifecycle.

Designing **modular and flexible systems** enables easier model replacement and reduces the risk of vendor lock-in and costly re-engineering.

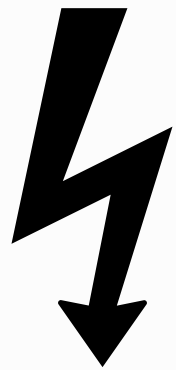
Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: Challenges of Classical AI through a Sustainability Lens

Environmental sustainability

What about the energy cost of training (periodic) and inference (continuous)?

Environmental sustainability is somehow strictly dependent on costs (and viceversa). Indeed, the root causes of poor sustainability are large-scale training, continuous inference at scale, and redundant experimentation.



Energy Consumption. Measured in kilowatt-hours (kWh), it represents the total electricity used by an AI system.

Carbon Emissions. Measured in kilograms of CO₂ equivalent (kgCO₂e), it represents the environmental impact of energy usage.

Compute Usage. Measured in GPU/TPU hours or FLOPs, it represents the computational effort.

Every API call, every model training instance, every prediction consumes energy.

Note that some environmental sustainability metrics, like carbon emissions, depend on **energy source** (renewable vs fossil): this means that models may have a different footprint depending on location.

Interestingly, energy consumption translates directly into costs: therefore, companies may be more proactive in addressing environmental issues when they also impact operational expenses.

Sustainability: Challenges of Classical AI through a Sustainability Lens

Environmental sustainability

What about the energy cost of training (periodic) and inference (continuous)?

Most practitioners tend to perceive environmental sustainability as a second-class concern, which companies often do not prioritize directly.

However, companies do care about costs, which may indirectly drive more sustainable practices.

Environmental sustainability is a good example of how adopting different terminology, e.g., ‘sustainability debt’, can influence how practitioners perceive and address the problem.

When asked about what they practically do to achieve sustainability goals, practitioners often refer to engineering and organizational practices that may indirectly contribute to sustainability, for example by improving efficiency, reducing resource usage, or optimizing system performance.

Sustainability of Machine Learning-Enabled Systems: The Machine Learning Practitioner’s Perspective

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STEFANO LAMBIASE, University of Salerno, Italy
FABIANO PECORELLI, Pegaso University, Italy
WILLEM-JAN VAN DEN HEUVEL, Tilburg University, Netherlands
FILOMENA FERRUCCI, University of Salerno, Italy
FABIO PALOMBA, University of Salerno, Italy

Software sustainability is a key multifaceted non-functional requirement that encompasses environmental, social, and economic concerns, yet its integration into the development of Machine Learning (ML)-enabled systems remains an open challenge. While previous research has explored high-level sustainability principles and policy recommendations, limited empirical evidence exists on how sustainability is practically managed in ML workflows. Existing studies predominantly focus on environmental sustainability, e.g., carbon footprint reduction, while missing *the broader spectrum of sustainability dimensions and the challenges practitioners face in real-world settings*. To address this gap, we conduct an empirical study to characterize sustainability in ML-enabled systems from a practitioner’s perspective. We investigate (1) how ML engineers perceive and describe sustainability, (2) the software engineering practices they adopt to support it, and (3) the key challenges hindering its adoption. We first perform a qualitative analysis based on interviews with eight experienced ML engineers, followed by a large-scale quantitative survey with 203 ML practitioners. Our key findings reveal a significant disconnection between sustainability awareness and its systematic implementation, highlighting the need for more structured guidelines, measurement frameworks, and regulatory support.

CCS Concepts: • **Software and its engineering** → **Software design engineering**; **Extra-functional properties**.

Additional Key Words and Phrases: Software Sustainability, Machine Learning-Enabled Systems, Empirical Software Engineering, Mixed-Method Research, Software Engineering for Artificial Intelligence

ACM Reference Format:
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Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: Challenges of Classical AI through a Sustainability Lens

Environmental sustainability

What about the energy cost of training (periodic) and inference (continuous)?

Empirical Software Engineering manuscript No.
(will be inserted by the editor)

Green Architectural Tactics in ML-enabled Systems: An LLM-based Repository Mining Study

Vincenzo De Martino · Silverio
Martínez-Fernández · Fabio Palomba

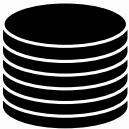
Received: date / Accepted: date

Abstract Context: The increasing adoption of machine learning (ML) and artificial intelligence (AI) technologies raises growing concerns about their environmental sustainability. Developing and deploying ML-enabled systems is computationally intensive, particularly during training and inference. Green AI has emerged to address these issues by promoting efficiency without sacrificing accuracy. While prior research has proposed catalogs of sustainable practices (i.e., green tactics), there remains limited understanding of their adoption in practice and whether additional, undocumented tactics exist. **Objective:** This study aims to investigate the extent to which existing sustainable practices are implemented in real-world ML-enabled systems and to identify previously undocumented practices that support environmental sustainability. **Method:** We conduct a mining software repository study on 205 open-source ML projects on GrrHub. To support our analysis, we design a novel mechanism based on large language models (LLMs) capable of identifying both known and new sustainable practices from code repositories. **Results:** Our findings confirm that green tactics reported in the literature are used in practice, although adoption rates vary. Furthermore, our LLM-based approach reveals nine previously undocumented sustainable practices. Each tactic is supported with code examples to aid adoption and integration. **Conclusions:** We finally provide insights for practitioners seeking to reduce the environmental impact of ML-enabled systems and offer a foundation for future research in automating the detection and adoption of sustainable practices.

Keywords Green AI · Machine Learning-Enabled Systems · Software Sustainability · Empirical Software Engineering

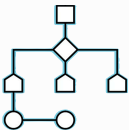
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Data

Reduce number of data features. Reducing the number of features can improve energy efficiency while maintaining accuracy.



Algorithm

Use built-in library functions. Built-in functions are usually optimized and well-tested, improving energy efficiency compared to custom functions.



Model

Energy-aware pruning. For each layer of a CNN model, pruning removes the layers having the highest energy consumption.



Deployment

Use computation partitioning. Process of dividing a CNN's computations between a mobile client and a cloud server.



Management

Consider Model Reproducibility. This ensures that ML models can be reproduced using the same conditions and configurations.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: Challenges of Classical AI through a Sustainability Lens

Environmental sustainability

What about the energy cost of training (periodic) and inference (continuous)?

Most green AI strategies can be implemented with just a few lines of code.

However, not all sustainability decisions happen in code. Some tactics must be addressed earlier, during system design or data analysis. This reminds us that software system development is more than just code: they are shaped by architecture, data, infrastructure, and human decisions.

Environmental sustainability requires collaboration between software engineers, data scientists, and other stakeholders. Sustainability is a multidisciplinary stratified effort that involves technical, environmental, and ethical considerations.

The lack of government incentives (both financial and regulatory) acts as a barrier to the development and widespread adoption of sustainable solutions.

The absence of purpose-built frameworks limits practitioners' ability to systematically embed sustainability considerations into development processes.

As a matter of fact, environmental sustainability is likely the dimension that is least explicitly considered and systematically addressed in current software engineering practices. This is partly because its impact is less immediate and less directly observable compared to other dimensions such as cost or performance.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: Challenges of Classical AI through a Sustainability Lens

Individual sustainability

Focus on the well-being of humans as individuals. This includes mental and physical well-being, education, self-respect, etc.

Similarly to social sustainability, this dimension refers to multiple aspects of human experience and well-being that are affected by AI systems.

Cognitive Impact. The effect of an AI system on users' ability to think, reason, learn, and make informed decisions over time.

Autonomy and Agency. The extent to which users retain control over decisions and actions when interacting with an AI system.

Well-being. The impact of an AI system on users' mental, emotional, and overall personal state during and after interaction.

Skill Development. The degree to which an AI system supports or hinders the growth and preservation of users' knowledge and abilities over time.

In other terms, AI systems do not just solve tasks: they shape how people think, learn, and act over time. As a consequence, a system that is efficient in the short term may still be unsustainable if it degrades human capabilities in the long run.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: Challenges of Classical AI through a Sustainability Lens

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Similarly to social sustainability, this dimension refers to multiple aspects of human experience and well-being that are affected by AI systems.

From an organizational perspective, companies should promote **training and awareness initiatives** to ensure that users engage with AI systems critically and responsibly.

This is closely related to the definition of human-centered policies, which aim to preserve users' autonomy, well-being, and long-term capabilities.

From an engineering perspective, developers should design AI systems that incorporate **human-in-the-loop mechanisms**, **explainability features**, and **adaptive interactions** to preserve user capabilities over time.

This is another example of how **sustainability emerges from the interplay of different roles**: HCI practices such as interaction design, explainable interfaces, control mechanisms, and learning-oriented feedback are essential to ensure that AI systems augment rather than replace human capabilities.

We will not elaborate further on these aspects here; they are covered in more detail in the 'Sustainable Software Engineering' course.

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Autonomy and Agency. The degree of control and decision-making when interacting with an AI system.

Well-being. The impact of AI on the mental and physical state during and after interaction.

Skill Development. The degree of skill acquisition and preservation of users' knowledge.

In other terms, AI systems must be designed to be sustainable and act over time. As a consequence, AI systems that are still be unsustainable if it does not.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: Challenges of Classical AI through a Sustainability Lens

Technical sustainability

What about model maintenance? What about the scalability of serving infrastructure?

Technical sustainability concerns the ability of AI systems to remain maintainable, reliable, and evolvable over time.

Essentially, what we discussed in this course can be framed under the lens of technical sustainability.

Key issues related to technical sustainability include maintainability, evolvability, reliability over time, reproducibility, and the management of technical debt and dependencies on data, models, and external infrastructures.

From an organizational perspective, organizations should adopt governance and lifecycle management practices that ensure AI systems can adapt to change while remaining reliable and maintainable over time.

This is basically what you will see in the ‘Software Project Management’ course, where you will learn how to structure processes and teams to keep software systems sustainable over time.

This is why we will not further elaborate on the organizational perspective of technical sustainability, as it is covered in more detail elsewhere.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: An Eye on GenAI-Enabled Systems

Generative AI does not introduce new sustainability dimensions: it amplifies all of them at scale. Indeed, as largely discussed throughout the course, it differs from traditional AI in scale, usage patterns, and deployment models.

Some key facts to consider when thinking at sustainability of GenAI-enabled systems:

- 1 As for training, we have solid estimates. For instance, training GPT-3 cost ~1,200–1,300 MWh electricity and ~500–550 tons CO₂. Just to give an idea, training a single large model can consume as much electricity as **100+ households in a year**.
- 2 As for inference, per-query numbers look small: ~0.24–0.3 Wh per request and ~0.03 g CO₂ per query. But scale changes everything. Inference accounts for 60–70% of total AI energy use. For popular GenAI models, we are talking about **trillions of queries/year**, which means that **inference can exceed training emissions within weeks or months!**
- 3 Considering the global impact, AI may reach 18–245 million tons CO₂ annually by 2035. The problem is not individual usage, but **aggregate scale**.

Despite these estimates, most companies do not disclose precise energy, carbon, or cost data. This is likely due to competitive concerns, the lack of standardized and enforceable reporting frameworks, and the complexity (sometimes opacity) of large-scale AI infrastructures.

In any case, the available data explains why sustainability is a big concern nowadays!

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: An Eye on GenAI-Enabled Systems



At the point that AI can play a key role in achieving the well-known United Nations Sustainable Development Goals.

For instance, Climate Action, Good Health and Well-being, Quality Education, and Reduced Inequalities can be advanced through and affected by AI technologies.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Before leaving... What Should You Remember?

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: What It Is About

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Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: What It Is About

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In addition to the concept of stratification, sustainability in software engineering is defined by other paradigms that help us determine its nature.

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(2) *Process Sustainability vs. Product Sustainability*: Software engineering affects sustainability both through the development process (e.g., how teams work) and the product (e.g., long-term usability). The practices are not always aligned: we can have a sustainable process but an unsustainable product and vice versa.

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Based on these paradigms, we can again rethink sustainability as an integrated concept throughout the software development lifecycle.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: Challenges of Classical AI through a Sustainability Lens

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Yet, equating social sustainability to fairness would be erroneous. Fairness is about whether a system treats people correctly. Social sustainability is about whether the system contributes to a **healthy society over time**.

Trust.

The degree to which individuals and society have confidence that an AI system behaves reliably, transparently, and in alignment with expected norms and values.

Impact on social structures.

The extent to which an AI system reshapes economic, professional, and institutional arrangements within society.

Behavioral and societal effects.

The influence of AI systems on individual behaviors and collective societal dynamics over time.

Inclusion and participation.

The degree to which diverse stakeholders are represented in data, involved in system design, and able to benefit from the system.

These dimensions determine whether an AI system contributes to a society that is fair, inclusive, and resilient over time.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: Challenges of Classical AI through a Sustainability Lens

Environmental sustainability	What about the energy cost of training (periodic) and inference (continuous)?
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Environmental sustainability is somehow strictly dependent on costs (and viceversa). Indeed, the root causes of poor sustainability are large-scale training, continuous inference at scale, and redundant experimentation.



Energy Consumption.

Measured in kilowatt-hours (kWh), it represents the total electricity used by an AI system.

Carbon Emissions.

Measured in kilograms of CO₂ equivalent (kgCO₂e), it represents the environmental impact of energy usage.

Compute Usage.

Measured in GPU/TPU hours or FLOPs, it represents the computational effort.

Every API call, every model training instance, every prediction consumes energy.

Note that some environmental sustainability metrics, like carbon emissions, depend on **energy source** (renewable vs fossil): this means that models may have a different footprint depending on location.

Interestingly, energy consumption translates directly into costs: therefore, companies may be more proactive in addressing environmental issues when they also impact operational expenses.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: Challenges of Classical AI through a Sustainability Lens

Individual sustainability	Focus on the well-being of humans as individuals. This includes mental and physical well-being, education, self-respect, etc.
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Similarly to social sustainability, this dimension refers to multiple aspects of human experience and well-being that are affected by AI systems.

Cognitive Impact.

The effect of an AI system on users' ability to think, reason, learn, and make informed decisions over time.

Autonomy and Agency.

The extent to which users retain control over decisions and actions when interacting with an AI system.

Well-being.

The impact of an AI system on users' mental, emotional, and overall personal state during and after interaction.

Skill Development.

The degree to which an AI system supports or hinders the growth and preservation of users' knowledge and abilities over time.

In other terms, AI systems do not just solve tasks: they shape how people think, learn, and act over time. As a consequence, a system that is efficient in the short term may still be unsustainable if it degrades human capabilities in the long run.

Software Engineering for Artificial Intelligence: Sustainability of AI Solutions

Sustainability: An Eye on GenAI-Enabled Systems

Generative AI does not introduce new sustainability dimensions: it amplifies all of them at scale. Indeed, as largely discussed throughout the course, it differs from traditional AI in scale, usage patterns, and deployment models.

Some key facts to consider when thinking at sustainability of GenAI-enabled systems:

1

As for training, we have solid estimates. For instance, training GPT-3 cost ~1,200–1,300 MWh electricity and ~500–550 tons CO₂. Just to give an idea, training a single large model can consume as much electricity as **100+ households in a year**.

2

As for inference, per-query numbers look small: ~0.24–0.3 Wh per request and ~0.03 g CO₂ per query. But scale changes everything. Inference accounts for 60–70% of total AI energy use. For popular GenAI models, we are talking about **trillions of queries/year**, which means that **inference can exceed training emissions within weeks or months!**

3

Considering the global impact, AI may reach 18–245 million tons CO₂ annually by 2035. The problem is not individual usage, but **aggregate scale**.

Despite these estimates, most companies do not disclose precise energy, carbon, or cost data. This is likely due to competitive concerns, the lack of standardized and enforceable reporting frameworks, and the complexity (sometimes opacity) of large-scale AI infrastructures.

In any case, the available data explains why sustainability is a big concern nowadays!



Software Engineering for Artificial Intelligence

Sustainability of AI-Enabled Systems

